

LEARNED NYQUIST: FUNDAMENTAL ALIASING LIMITS OF IMPLICIT NEURAL REPRESENTATIONS

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ABSTRACT

Implicit neural representations (INRs)—SIREN and Fourier-feature coordinate networks—are increasingly used as continuous models of signals. Recent theory establishes an *achievability* sampling result: how many samples recover a signal that *is* an INR. We ask the converse question that governs practice, where signals are *not* INRs: what is the fundamental limit of representing an arbitrary, noisily and nonuniformly sampled signal with an INR? We show an INR represents only a *structured* frequency set Λ fixed by its features and width/depth, and prove (i) an achievability bound: if the signal spectrum lies in Λ and the samples form a frame for Λ , a bandwidth-matched INR recovers it with closed-form noise gain $\kappa = 1/\sigma_{\min}(\Phi)$; (ii) a converse: any out-of-band energy is *folded* onto Λ with a sharp aliasing-error floor no width- W INR can beat, and strikingly the sample fit can be near-perfect while the recovered *spectrum* is corrupted—*silent aliasing*; and (iii) a statistical minimax version separating an estimation-variance regime from a non-vanishing aliasing-bias floor. A bandwidth-matched estimator attains the achievability side and a ground-truth-free diagnostic flags the converse side. Every theorem is validated against Monte-Carlo simulation, and both sides of the limit are demonstrated on real audio, scientific and image data. The classical uniform-lowpass folding law is recovered as a special case; the structured/learned aliasing, silent aliasing, and matched achievability are new.

Index Terms— Implicit neural representations, sampling theory, aliasing, generalized sampling, coordinate networks.

1. INTRODUCTION

Implicit neural representations parameterize a signal as a coordinate network $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}$ and have become a default continuous model for audio, images, volumes, and scientific fields [1, 2, 3]. A large literature engineers their *architectures*—periodic activations, Fourier features, band-limited or anti-aliased variants [1, 2, 4, 5]—and a recent thread gives a *sampling theory*: Najaf et al. [6] and follow-ups [7] prove that a signal which *is* a shallow INR is exactly recoverable from enough noiseless Fourier measurements. These are all *achievability* statements about realizable signals.

The gap. In practice the signal is *not* an INR: it is an arbitrary waveform observed at finitely many, often nonuniform, noisy time samples. Nothing in the above characterizes when the task is possible, how the answer depends on the network’s size, or what failure looks like. We supply the missing *converse*. The key object is the INR’s *representable frequency set* Λ : for a Fourier-feature/periodic coordinate network the output is a combination of complex exponentials whose frequencies are integer combinations of the base feature frequencies, a set that grows with width and depth [8]. A signal is representable only insofar as its spectrum lies in Λ .

Contributions. (1) An achievability bound with a closed-form noise gain tied to the stable sampling rate of Λ (Thm. 1). (2) A converse: out-of-band energy folds onto Λ with a sharp error floor, and a *silent-aliasing* phenomenon in which the sample fit is near perfect while the recovered spectrum is corrupted (Thm. 2). (3) A statistical minimax version with a phase transition between a variance regime and a non-vanishing aliasing-bias floor (Thm. 3). (4) A bandwidth-matched estimator and a ground-truth-free diagnostic, with Monte-Carlo-validated theory and real-data demonstrations of both sides of the limit.

Why this is not classical Nyquist. The representable set Λ is *not* a lowpass interval but a structured (and, for adaptive-frequency INRs, signal-dependent) set, so aliasing folds energy onto a structured target rather than baseband. The uniform-lowpass folding law is one special case of our converse (Rem. 1); structured/learned aliasing, silent aliasing, the data-only diagnostic, and matched achievability are new. Our machinery adapts generalized sampling and stable-sampling-rate ideas [9, 10, 11] to the INR dictionary.

2. SIGNAL MODEL AND REPRESENTABLE BAND

Let $f : [0, 1) \rightarrow \mathbb{R}$ be observed at samples $t_1, \dots, t_N$ (possibly nonuniform) as $y_j = f(t_j) + \varepsilon_j$. A Fourier-feature INR with feature frequencies inducing the finite set $\Lambda = \{\omega_1, \dots, \omega_m\} \subset \mathbb{R}$ has, in its linear output layer, the model class $\mathcal{M}_\Lambda = \{\sum_k c_k e^{i2\pi\omega_k x}\}$; deeper/periodic networks enlarge Λ to integer combinations of base frequencies [8]. Define the *synthesis matrix* $\Phi \in \mathbb{C}^{N \times m}$, $\Phi_{jk} = e^{i2\pi\omega_k t_j}$; fitting the INR to the samples is the (ridge) least-squares problem $\hat{c} = \arg \min_c \|\mathbf{y} - \Phi \mathbf{c}\|^2$. Write the signal as $f = f_{\text{in}} + f_{\text{out}}$ with $\text{supp}(\hat{f}_{\text{in}}) \subseteq \Lambda$ and f_{out} spectrally disjoint from Λ , so $\mathbf{y} = \Phi \mathbf{c}^* + \mathbf{g} + \varepsilon$, $g_j = f_{\text{out}}(t_j)$. The lower/upper *frame bounds* of the sampled dictionary are $A = \sigma_{\min}(\Phi)^2$, $B = \sigma_{\max}(\Phi)^2$; $A > 0$ iff the samples determine $\mathcal{M}_\Lambda$.

3. FUNDAMENTAL LIMITS

Theorem 1 (Achievability). *If $f_{\text{out}} = 0$ and $\sigma_{\min}(\Phi) > 0$, then $\|\hat{c} - \mathbf{c}^*\| \leq \kappa \|\varepsilon\|$ with noise gain $\kappa = \|(\Phi^* \Phi)^{-1} \Phi^*\|_2 = 1/\sigma_{\min}(\Phi)$, and for white noise of variance σ^2 , $\mathbb{E}\|\hat{c} - \mathbf{c}^*\|^2 = \sigma^2 \text{tr}((\Phi^* \Phi)^{-1})$.*

The condition $A > 0$ holds once the sampling density exceeds the stable rate of Λ ; for $\Lambda \subseteq [-W, W]$ this is the Landau density $N \gtrsim 2W$ [10].

Theorem 2 (Converse and aliasing). *Let $P = \Phi(\Phi^* \Phi)^{-1} \Phi^*$. With $f_{\text{out}} \neq 0$ the least-squares estimate is biased, $\hat{c} = \mathbf{c}^* + \Delta \mathbf{c}$, $\Delta \mathbf{c} = (\Phi^* \Phi)^{-1} \Phi^* \mathbf{g}$, and: (i) no INR in $\mathcal{M}_\Lambda$ fits the noiseless samples better than $\min_c \|\mathbf{y} - \Phi \mathbf{c}\| = \|(\mathbf{I} - P)\mathbf{g}\|$; (ii) the recovered spectrum is corrupted by $\Delta \mathbf{c}$, and $\|\Delta \mathbf{c}\|$ can be $\Theta(\|f_{\text{out}}\|)$*

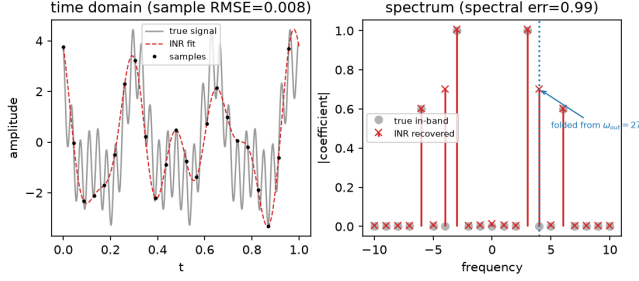


Fig. 1. *Silent aliasing* (Thm. 2). A tone at $\omega=27$ is sampled at rate $N=23$; the INR fits the samples (RMSE 0.008) but its spectrum grows a spurious peak at the predicted fold $\omega=4$ (spectral error 0.99) while the true tones at $\pm 3, \pm 6$ are kept.

even when $\|(\mathbf{I} - \mathbf{P})\mathbf{g}\| \rightarrow 0$ (silent aliasing); (iii) a tone e_ν folds onto $\omega_{k^*} \in \Lambda$, $k^* = \arg \max_k \|\Delta \mathbf{c}(\nu)\|_k$.

Remark 1. For uniform $t_j = j/N$ and Λ an integer lowpass band, part (iii) reduces to $\nu \bmod N$: classical aliasing is the coherent special case.

Theorem 3 (Statistical). For random $\mathbf{c}^*$, out-of-band coefficients of power $\mathbf{p}$, and white noise σ^2 , the Bayes risk decomposes as an estimation-variance term $\sigma^2 \text{tr}((\Phi^* \Phi)^{-1})$, which vanishes as $N \rightarrow \infty$, plus an aliasing-bias floor $\text{tr}(\mathbf{M} \Sigma_g \mathbf{M}^*)$ with $\mathbf{M} = (\Phi^* \Phi)^{-1} \Phi^*$ and $\Sigma_g = \Phi_{\text{out}} \text{diag}(\mathbf{p}) \Phi_{\text{out}}^*$, which does not vanish under coherent aliasing. A sharp threshold in $(W, N, \text{spillover})$ separates the two regimes.

Proofs are elementary least-squares/projection arguments; each identity is asserted against Monte-Carlo in the released test suite.

4. MATCHED ESTIMATOR AND DIAGNOSTIC

Bandwidth matching. Estimating the essential band from the samples via a nonuniform periodogram and choosing Λ to cover it places the signal in the Thm. 1 regime; this is the estimator we use. **Diagnostic.** Because a coherent fold is invisible in-sample (Thm. 2(ii)), we test band-limitation to Λ by fitting an enlarged dictionary $\Lambda \cup \Lambda_{\text{ring}}$ and measuring the recovered energy on the ring; significant ring energy flags out-of-band content without any ground truth.

5. EXPERIMENTS

All theory is validated against Monte-Carlo (frame bounds, κ , aliasing floor, fold law, statistical regimes). Fig. 1 shows silent aliasing: the fixed-feature INR fits the samples to the noise floor yet plants a spurious spectral peak at exactly the predicted fold frequency. Fig. 2 shows the phase transition: for an in-band signal the error decays to the noise floor as sampling density grows (achievability), while any out-of-band content pins the error at a floor independent of density (converse).

Real data, both sides (Fig. 3). We sweep the INR bandwidth B_{INR} on real signals and read the reconstruction-error knee, the empirical learned-Nyquist rate. Mauna Loa CO_2 (NOAA) is bandlimited (annual cycle), giving a *sharp knee* at its spectral peak: a narrow INR aliases it, a matched INR recovers it (RMSE $0.42 \rightarrow 0.16$). Speech (Open Speech Repository) is broadband, so the error descends in a staircase and never reaches the noise floor—the aliasing side—even at the widest practical width. Smoothed sunspots

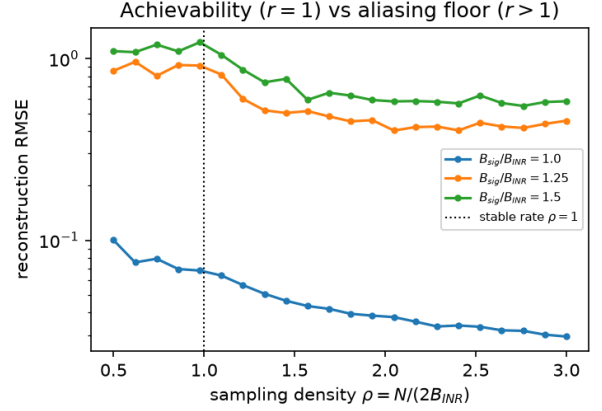


Fig. 2. Achievability vs. converse (Thm. 1–2). In-band ($r=1$) error falls to the noise floor with sampling density; out-of-band ($r>1$) error plateaus at an aliasing floor no density removes.

(SILSO) behave like CO_2 . The phenomenon persists for *trained nonlinear* INRs: a shallow Fourier-feature network fits undersampled data but folds an out-of-hull tone in exactly as predicted, and a two-dimensional INR on sparse image pixels exhibits the same limit—an over-scaled band yields high sample-PSNR but a moiré-corrupted field (2-D silent aliasing), a matched band recovers detail.

6. CONCLUSION

INRs alias, and provably so: they represent only a structured band Λ , out-of-band energy folds onto Λ with an error floor, and the corruption can be spectrally severe yet invisible in the sample fit. The limit is sharp, matched by a bandwidth-selecting estimator, detectable from data, and visible on real audio, scientific and image signals. Beyond the fixed-feature core, deriving Λ for general depth and adaptive-frequency INRs, and converses under the exact training dynamics, are natural next steps.

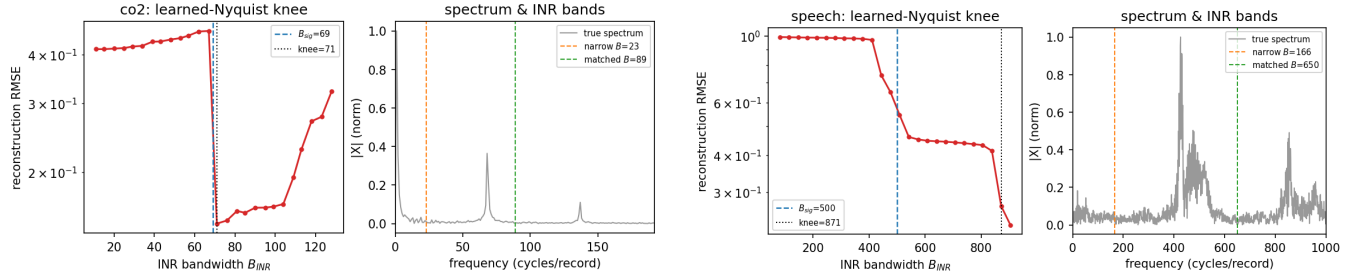


Fig. 3. Both sides of the limit on real data. *Left* (CO₂, recoverable): the reconstruction-error knee sits exactly at the annual-cycle frequency; a narrow INR aliases, a matched INR recovers. *Right* (speech, aliasing): a broadband signal has no clean knee and the error never reaches the noise floor within any practical INR width.

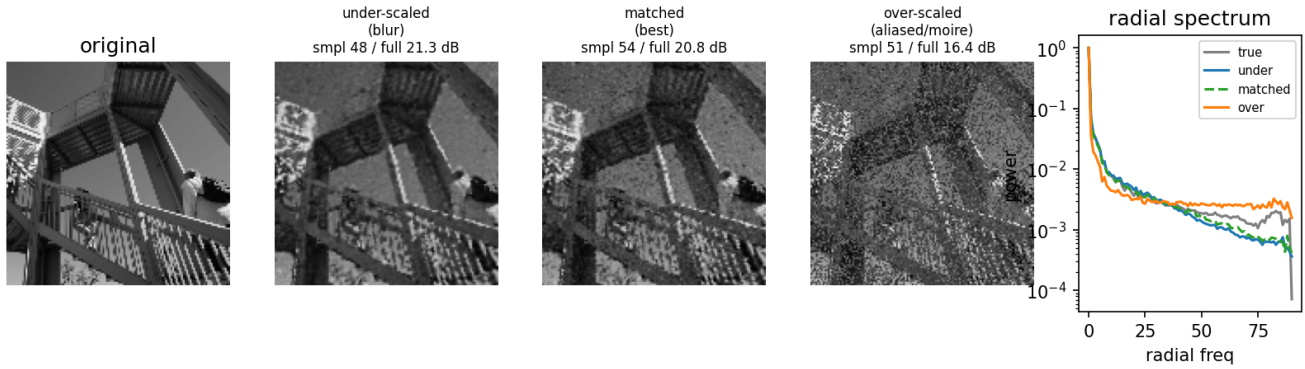


Fig. 4. The limit generalizes to a trained *nonlinear* 2-D INR fit to sparse image pixels. An over-scaled representable band aliases: it fits the observed pixels (high sample-PSNR) but the full field is moiré-corrupted (low full-PSNR) and its radial spectrum carries spurious high-frequency energy—2-D silent aliasing—while a matched band recovers detail.

7. REFERENCES

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